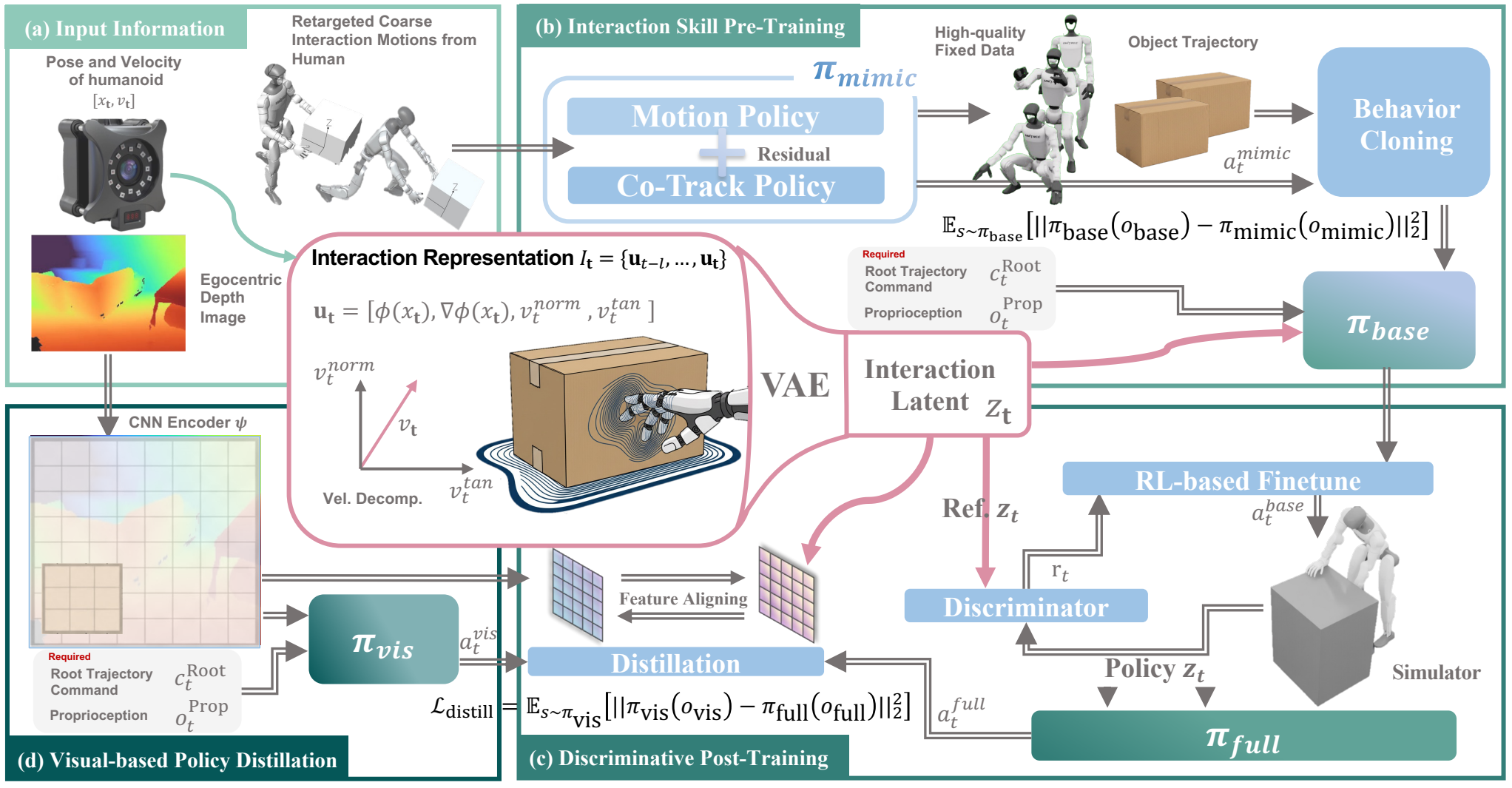
(a) Input Information
Pose and Velocity of humanoid
$[x_t, v_t]$
Retargeted Coarse Interaction Motions from Human
Egocentric Depth Image
(b) Interaction Skill Pre-Training
π_{mimic}
Motion Policy
Residual
Co-Track Policy
High-quality Fixed Data
Object Trajectory
a_t^{mimic}
Behavior Cloning
$\mathbb{E}_{s\sim\pi_{\text{base}}}[||\pi_{\text{base}}(o_{\text{base}}) - \pi_{\text{mimic}}(o_{\text{mimic}})||_2^2]$
Interaction Representation $I_t = \{\mathbf{u}_{t-l}, \dots, \mathbf{u}_t\}$
$\mathbf{u}_t = [\phi(x_t), \nabla\phi(x_t), v_t^{norm}, v_t^{tan}]$
v_t^{norm}
v_t
v_t^{tan}
Vel. Decomp.
VAE
Interaction Latent z_t
Required
Root Trajectory Command c_t^{Root}
Proprioception o_t^{Prop}
π_{base}
CNN Encoder ψ
Ref. z_t
RL-based Finetune
a_t^{base}
r_t
Discriminator
Feature Aligning
Distillation
π_{vis}
a_t^{vis}
Required
Root Trajectory Command c_t^{Root}
Proprioception o_t^{Prop}
$\mathcal{L}_{\text{distill}} = \mathbb{E}_{s\sim\pi_{\text{vis}}}[||\pi_{\text{vis}}(o_{\text{vis}}) - \pi_{\text{full}}(o_{\text{full}})||_2^2]$
Policy z_t
Simulator
a_t^{full}
π_{full}
(d) Visual-based Policy Distillation
(c) Discriminative Post-Training